CSE 216 Formal Methods
Fall 2026 Daniel Fremont

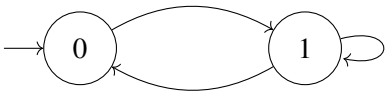
HW 5

Due November 3 at 3:00 pm

(1 question, 45 points total)

1. (45 pts.) Transition systems

Consider the following closed transition system M:



(a) What are all runs of M of length 3? (You can write them as triples of states.)

Answer: 010

011

(b) Let φ be the trace property "state 0 is visited infinitely many times". Does $M \models \varphi$? Why or why not?

Answer: no. there exists a trace that goes $(0,1,1,1,1\ldots)$ with only 1s that does not have state 0 infinitely many times.

(c) Let's run the BDD-based algorithm for computing the reachable states of M. First, write a BDD for the function $S_0(s)$ representing the initial states, which will also be the BDD $R_0(s)$ of states reachable in at most 0 steps.

Answer: since there are only 2 states, a single bit is enough to represent the states.

$$S_0(s) = \neg s \tag{1}$$

(d) Write a BDD for the function $\delta(s,s')$ representing the transition relation.

Answer:

$$\delta(s,s') = (\neg s \wedge s') \vee (s \wedge \neg s') \vee (s \wedge s') \tag{2}$$

(e) Write a BDD for the function $R_0(s) \wedge \delta(s,s')$.

Answer:

$$R_0(s) \wedge \delta(s,s') = \neg s \wedge (s \vee s') = (\neg s \wedge s) \vee (\neg s \wedge s') = 0 \vee (\neg s \wedge s') = \neg s \wedge s' \tag{3}$$

(f) Write a BDD for the function $\exists s.(S_0(s) \wedge \delta(s,s'))$ (which should only depend on s').

Answer:

$$\exists s.(\neg s \wedge s') = (\neg 0 \wedge s') \vee (\neg 1 \wedge s') = (1 \wedge s') \vee (0 \wedge s') = s' \tag{4}$$

(g) Write a BDD for the function $R_1(s') = R_0(s) \vee \exists s.(S_0(s) \wedge \delta(s,s'))$.

Answer:

$$R_1(s') = \neg s' \vee s' = 1 \text{ (True)} \tag{5}$$

(h) What will happen in the next iteration of the algorithm?

Answer: A fixpoint is reached because R_1 covers all states (0 and 1). The next iteration will yield $R_2 = R_1$, so the algorithm terminates with all states marked as reachable.